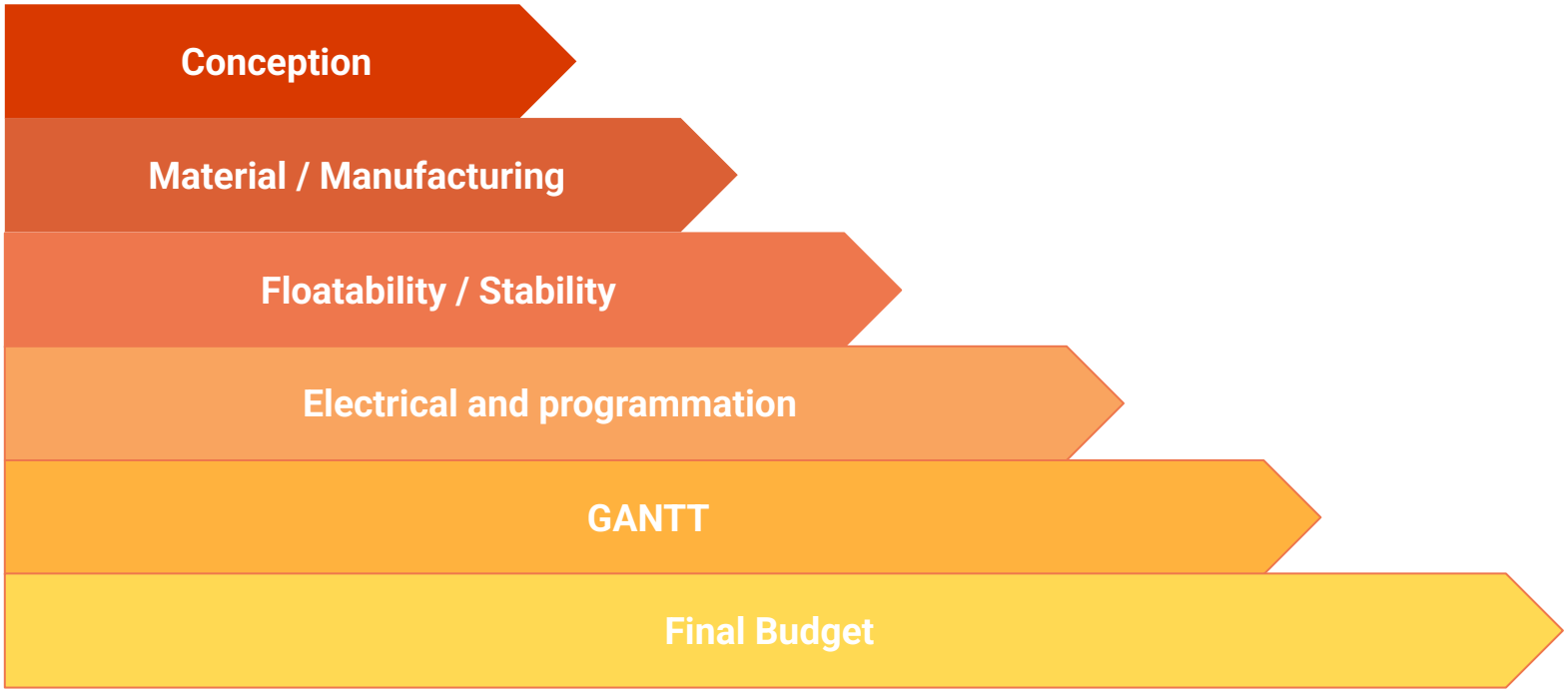
Leisure Boat

P5 group

William JUNIOR-ABO, Lucas LE GOFF, Theo SCHLOESSER-GRIMAUD, Lilou PICHON

Table of contents



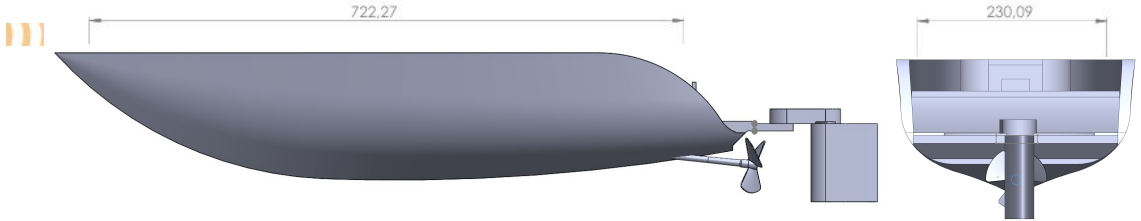
Overview

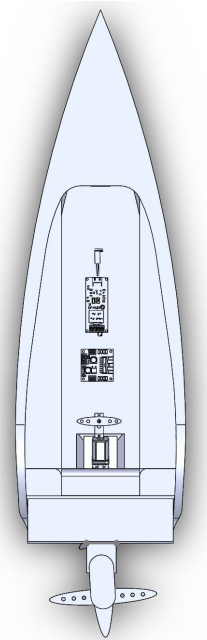
Length : 72 cm

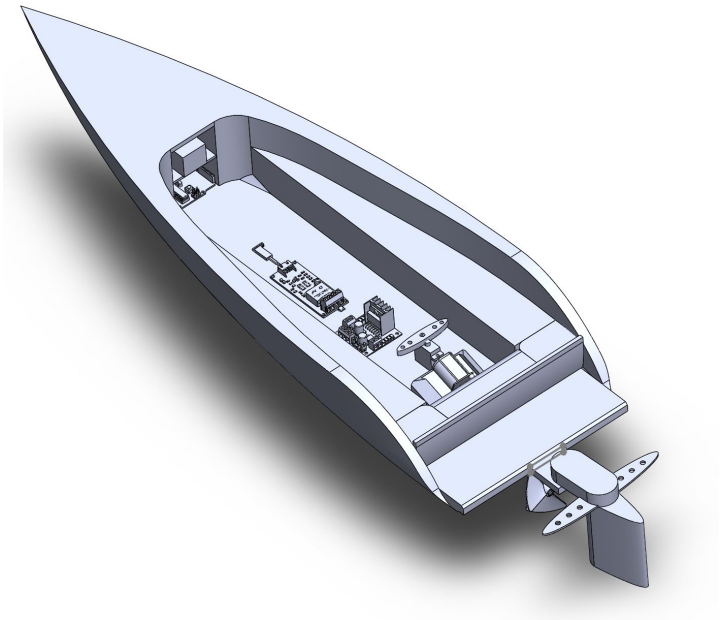
Width : 22 cm

Height : 129 cm

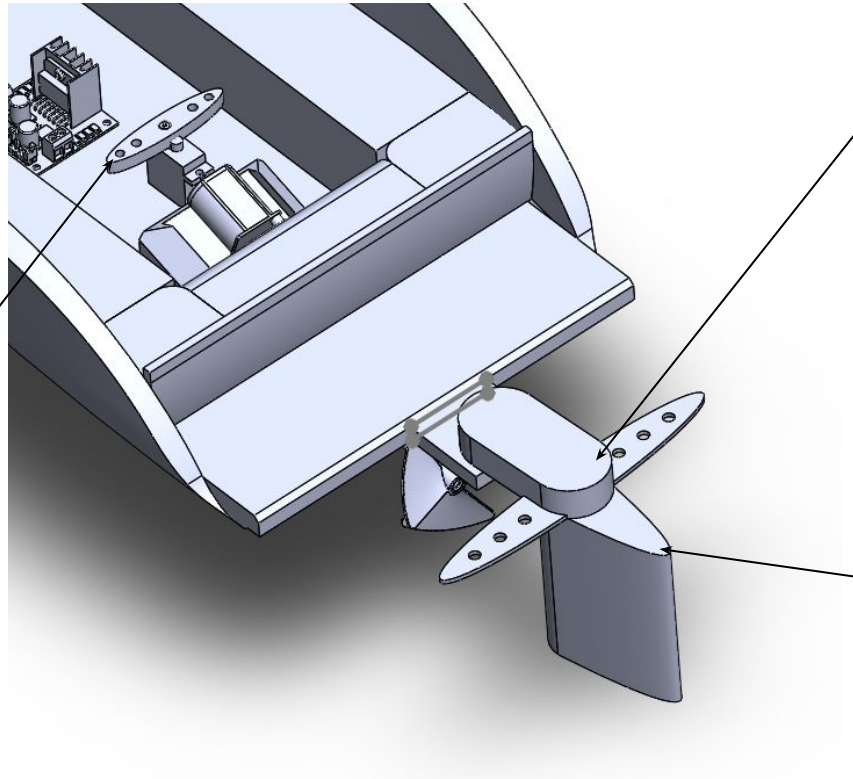
Weight : 6 kg







Rudder CAD



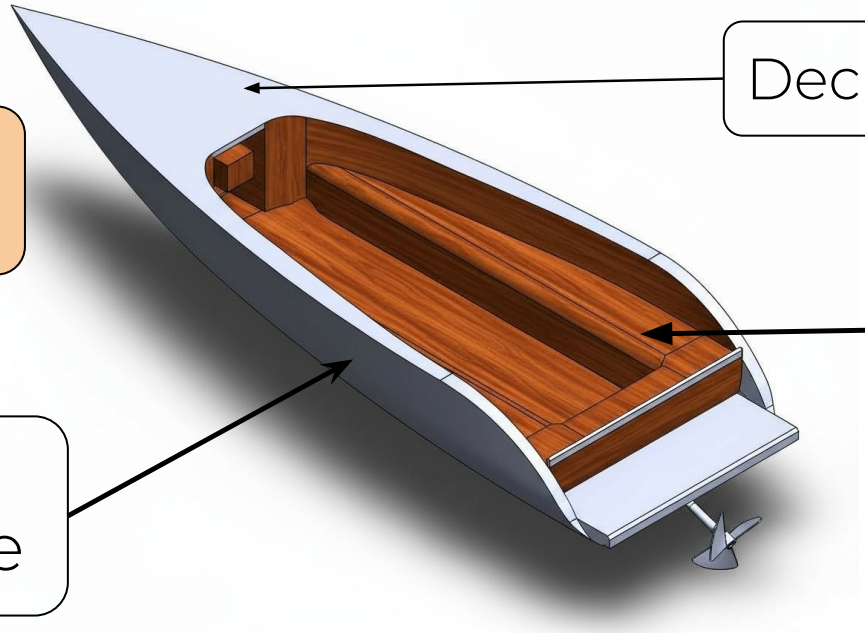
Link to the servo-motor

Pivot fixation

PLA printed piece

Manufacturing Plan

Adhesive bonding



Deck in composite

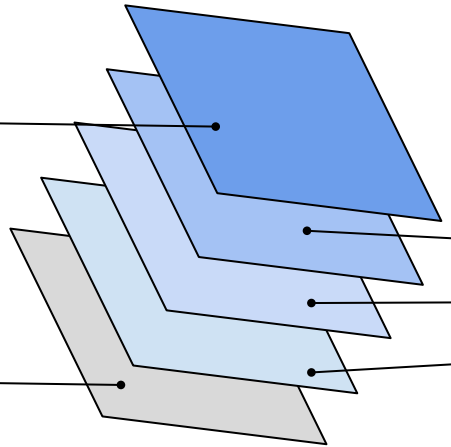
Bench in wood

Hull in composite

Glass Fiber Composite

300g/m optimal last layer for closing draping

100g optimal first layer for draping curves



Adding mastic then gel coat for finiton

300/450g trade-off between thickness and mass (samples test)

⚠ Alternate with polyester resin layers

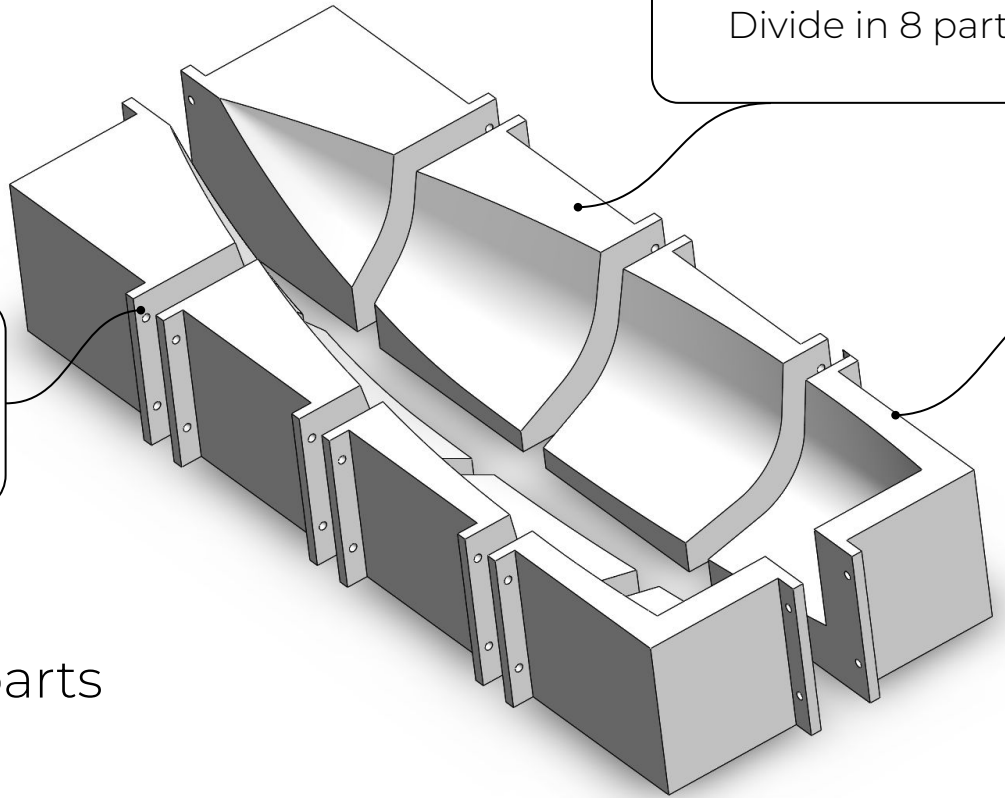
Pre-Molding wax dried layers

Hull Mold



M6

Screw assembly with 2 7,5mm holes per parts



Divide in 8 parts

PLA



For 4kg, 80€

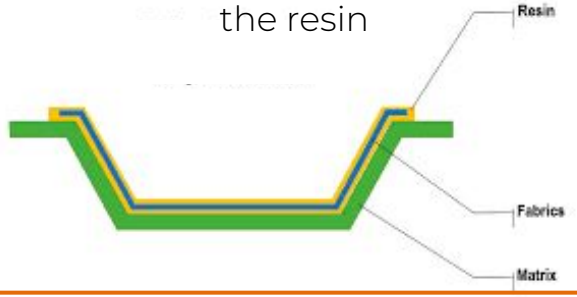
Risk: holes, gap parts

Hull manufacturing process

Dimension the fiber glass sheets
Prepare the mix (resin + hardener)



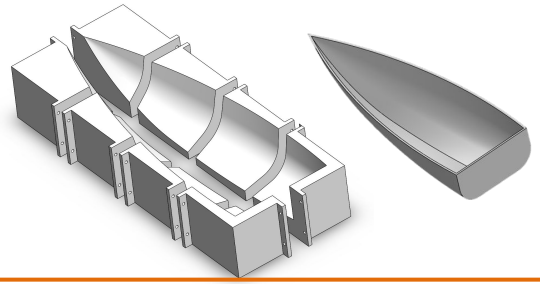
Lay the fiber glass in the mold and apply the resin



Let dry for 24h



Separate the 8 parts of the mold

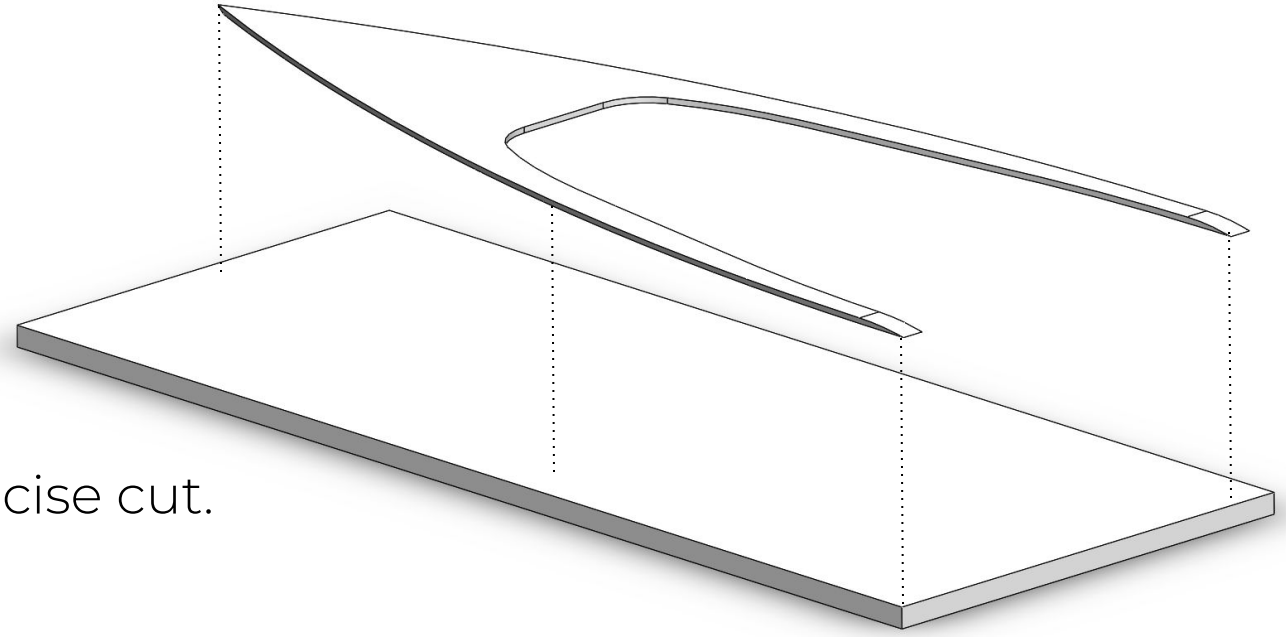


Sanding + surface protection



Deck manufacturing process

We will make the deck by tracing the form on a glass fiber rectangle and then we cut it.



Risk: unprecise cut.

Fiber



Glass Fiber MAT

Fiber Mass = Hull surface area* g/m³
= 0.103889 * 1450
= 151 g

1,37 €

Resin



Polyester resin



*

Resin quantity = Fiber Mass * 2.5
= 151*2.5
= 377 g

1 kg pot = 10 €

*catalyst (7.06€)

Gel Coat



Gel Coat + Catalyst

50g of gel coat for 7mL* of catalyst

= 18.74€

350g/m2 of gel coat

*2% of the gel coat quantity

Composite Concentration



Glass Fiber MAT

151g



Polyester resin

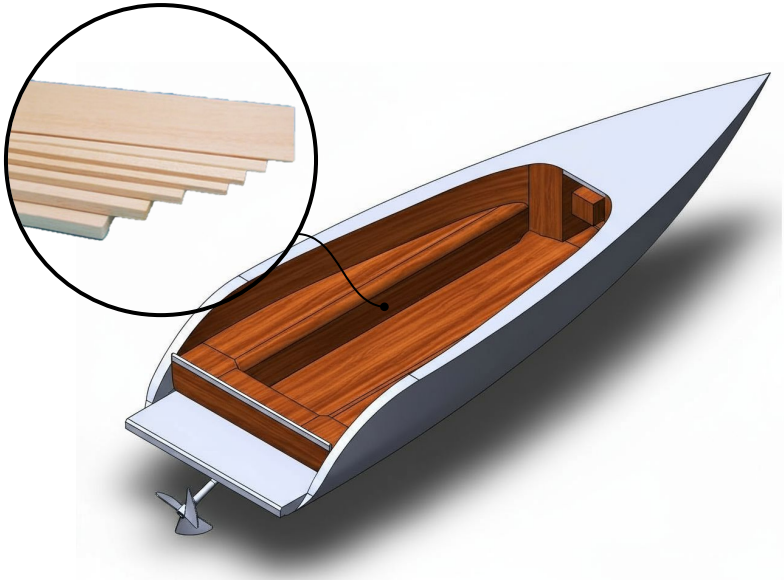
377g

Criterias	30% Fiberglass	50% Fiberglass
Stiffness / Strength	Good	Excellent
Shock Absorption	Better (more resin = more flexibility)	Lower
Ease of Manufacturing	Easy (ideal for hand lay-up)	Difficult (requires vacuum infusion or advanced methods)
Cost	Lower (less fiber used)	Higher (more fiber and complex process)

—> 30% fiberglass offers an optimum compromise between sufficient rigidity, ease of manufacturing

Interior Plan

Balsa board on the laser cutter

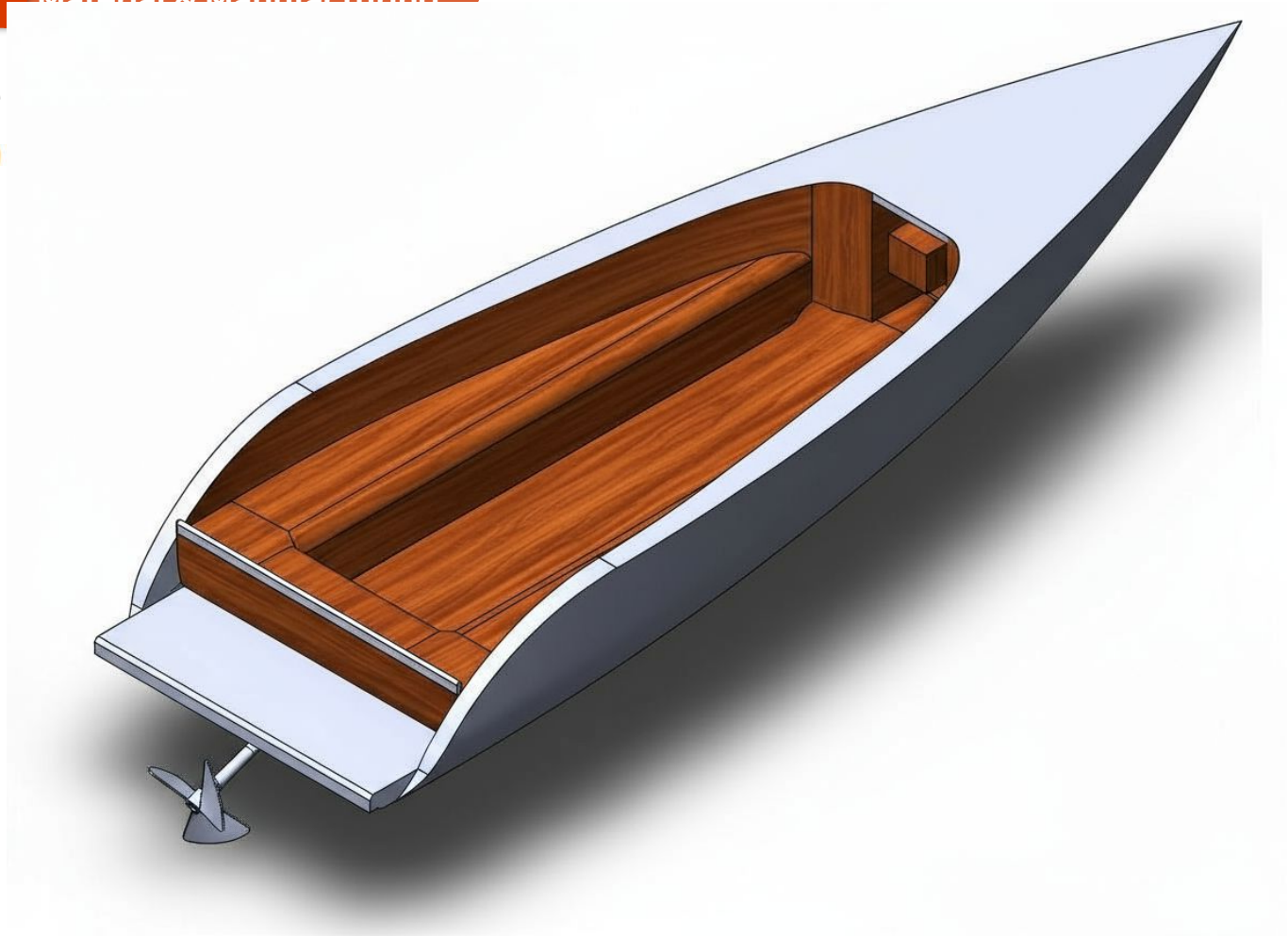


Wood part

- We will do this in multiple parts
- Then will we glue all the part between them
- Finally we will glue wood part with the deck

Make sure to make symmetrical part which fits good in the hull

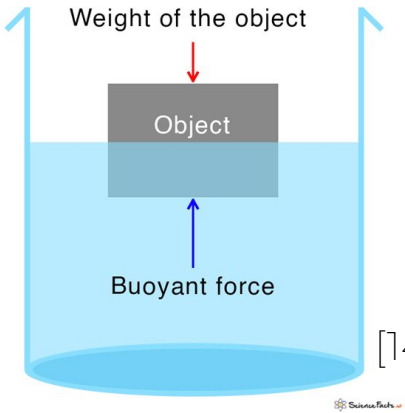
Inte



Does our boat float ?

"Any object fully or partially submerged in a fluid experiences an upward force equal to the weight of the fluid it displaces"

Archimedes Principle



[14]

BUOYANCY	
F buoyancy (N)	59,4
Weight (N)	58,8
Does it floats ?	YES !!

Weight = m x g

F buoyancy= ρ x V x g

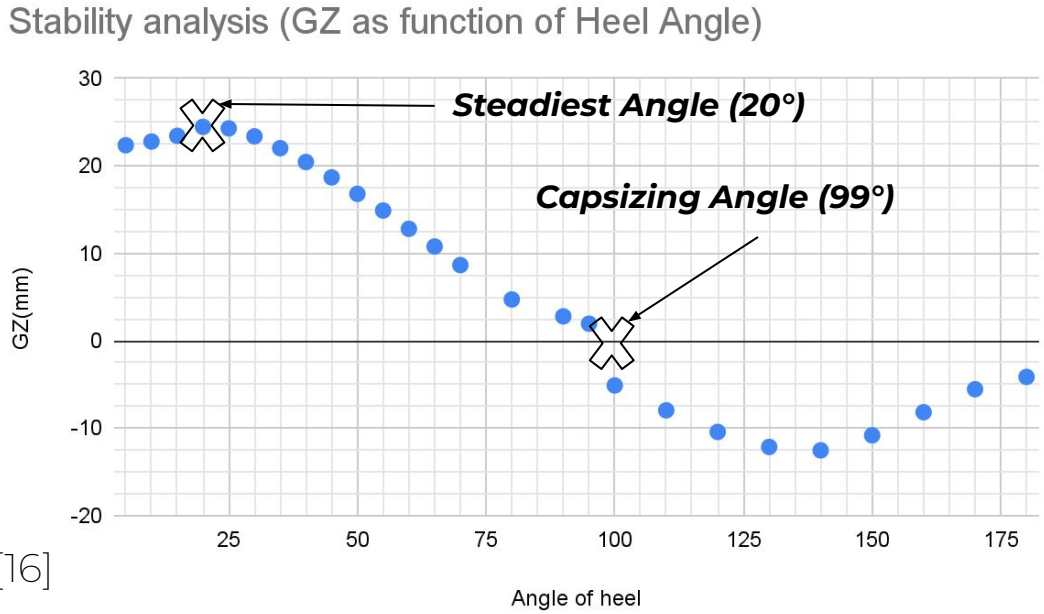
Boat mass = 6 kg

Gravitational acceleration = 9.81 m/s^2

Fluid water density = 1000 kg/m^3

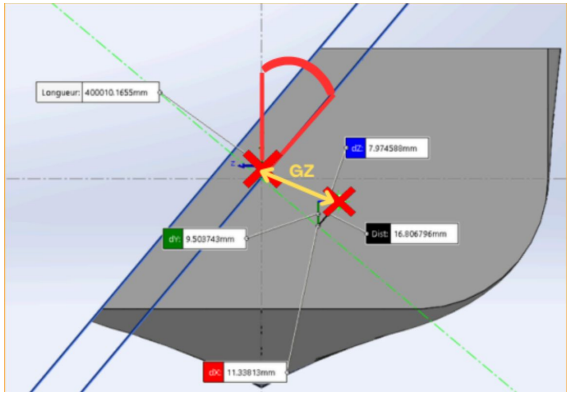
Volume displaced =0,006 m^3

Our results...



[16]

Biggest arm lever: steadiest
Smallest arm lever : capsizing point

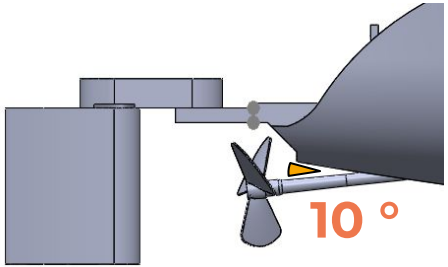


[17]

A SolidWorks simulation of the variation of tilt angles will give us significant points with measuring GZ distance (lever righting arm)

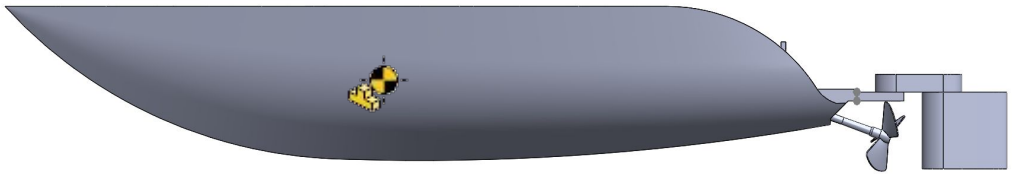
Stability control

Shaft Angle for ideal propulsion:

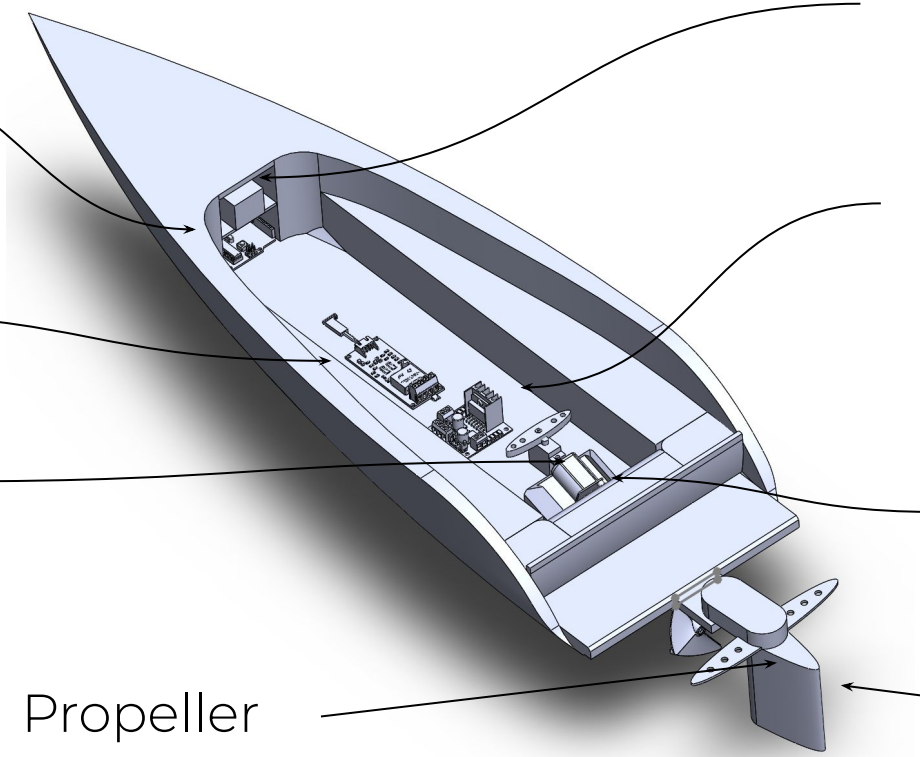


Load distribution between main component straight (battery and motor)

Center of gravity:



Component distribution



Propulsion

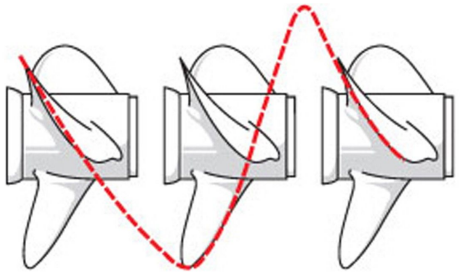


Propeller Characteristics



Brass

Propeller reference



Pitch providing forward travel to a prop

Number of blades	3 (Ideal balance between thrust & efficiency)
Diameter (75% of the draft)	0,75*7,8= 5,85 cm
Pitch (horizontal distance of one spin)	3,9 cm
Ae/Ao	65 %
Power (W)	100
Rotational speed	9500 rpm

DC Motor



Main characteristics:

Model Name: High Torque Motor 12V DC 775
Voltage: 12 Volts
Power: 100 Watts
Material: Copper, Metal
Weight: 100g
Rotation speed: 12,000 rpm
Angular speed: 995 rad/s
Dimensions : 98 * 66 * 42 mm
Current needed input: 10A

23.99 €

Motor current calculation



Power (Watts) = Voltage (V)×Current (A)

⇒Current (A) = Power (W) / Voltage (V) = 100/12 = **8.333A**

Why this choice?

- Dimensions or Sizing
- Its characteristics fits with the power needed by the propellers and more suitable for compact boats like ours.

Battery

Technical Specification

Battery Type:	LiFePO4	Voltage:	12V
Net Weight:	886g/31.25oz	Capacity:	8Ah
Size(L*W*H):	150*65*94mm	Cycle Life:	4000+



- Number of battery : **LiFePO4**
- Capacity : **8000 mAh (8AH)**
- Voltage: **12 V**
- Weight: **886g**
- Out put current: **10A Max**

Autonomy (hours) = Battery Capacity (Ah) /Motor Current Draw (A)

= 8/8.33 = 0.96h

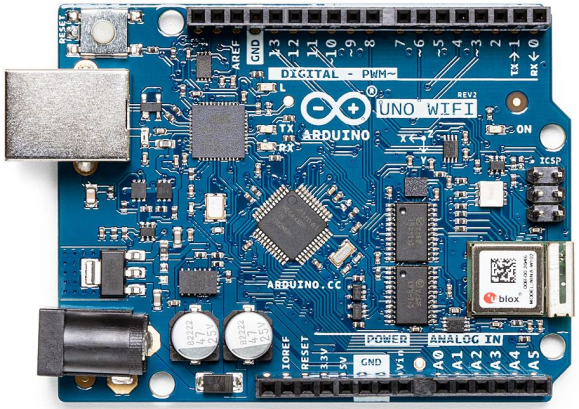
So **≈ 0.96 h ≈ 58 minutes** in an ideal world

39.99 €

34.44 €

TOTAL

Arduino card



- Model name: **UNO WiFi REV2**
- Connectivity technology: **USB**
- Included components: **UNO R2 Wifi card, and devices compatible with UART, SPI, I2C**
- Wireless communication standard: **Bluetooth**
- Weight: **25g**

46 €

80.44 €

TOTAL

H-Bridge



- Module model: Hailege BTS7960
- Logic voltage: 5V
- Drive voltage: 5V-27V
- Drive current: 43 A peak
- Ideal for: 12 V 775 motor class
- Dimensions: 43L x 43W x 27H millimeters

13.19 €

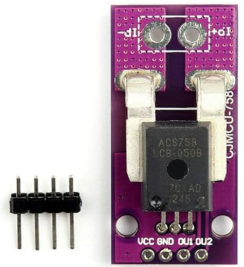
86.43 €

TOTAL

Sensors

Current

13,49 €

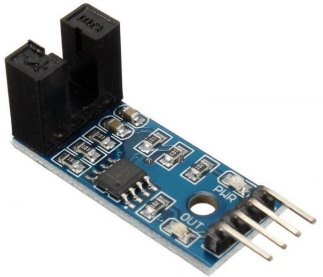


ACS758 ACS758LCB-050B-PFF-T 50A

- VCC: **3V - 5.5V**
- Measurement Range: **±50 A**
- Dimensions: **40 mm - 20 mm**
- Output: Analog voltage proportional to current (≈ 40 mV/A sensitivity)

Speed

6 €



LM393 Arduino

- - 2 outputs: 1 digital, 1 analogue
- - Power LED
- - Operating voltage: **3.3 to 5 VDC**
- - Weight: **8 g**
- - Dimensions: **32 x 14 x 7 mm**

95.38 €

TOTAL

Protectives component

Diode

0.99 €



- Voltage: **45 V**
- Max current: **15A - 20A**
- Weight: **1.3g**

Switch

10€



- Nominal voltage: **2.7 V to 16 V**
- Nominal current **10 A**
- Dimension: **17,78 × 12,7 × 3,30 mm**
- Weight: **0.6 g**

87.56 €

TOTAL

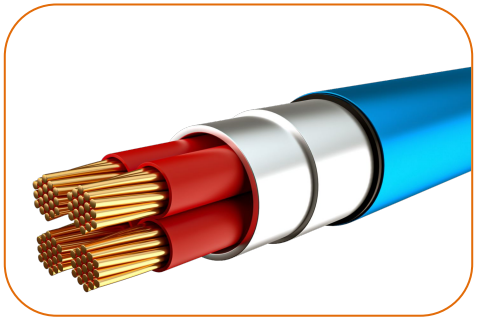
Security Norms

French Security Norms:

Our case is a **SELV** (Safety Extra-Low Voltage)

Wires gauge adapted
(2 - 3mm²)





Insulation sheath on wires



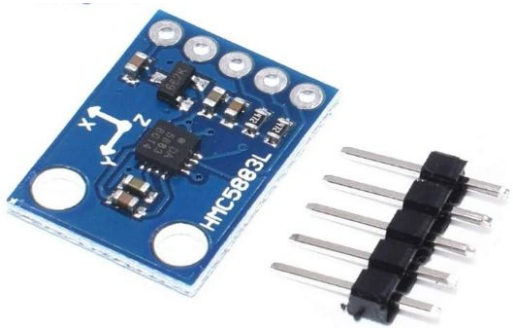
Limited current (using a diode)

Servo motor and Compass Magnetometer



- Model: SG90
- Power supply :5V
- Size: 3,3 x 3 x 1,2 cm;
- Speed Response: 0.3s
- Net weight: 14 g

4.99€



- Model: GY-273 QMC5883L
- Communication: I2C
- Measuring range: ± 1.3-8 gauss
- Power supply :3-5V
- Size: 1.8 x 1.3 cm
- Net weight:17 g

10.66€

111.03 €

TOTAL

Installation of the servo-motor

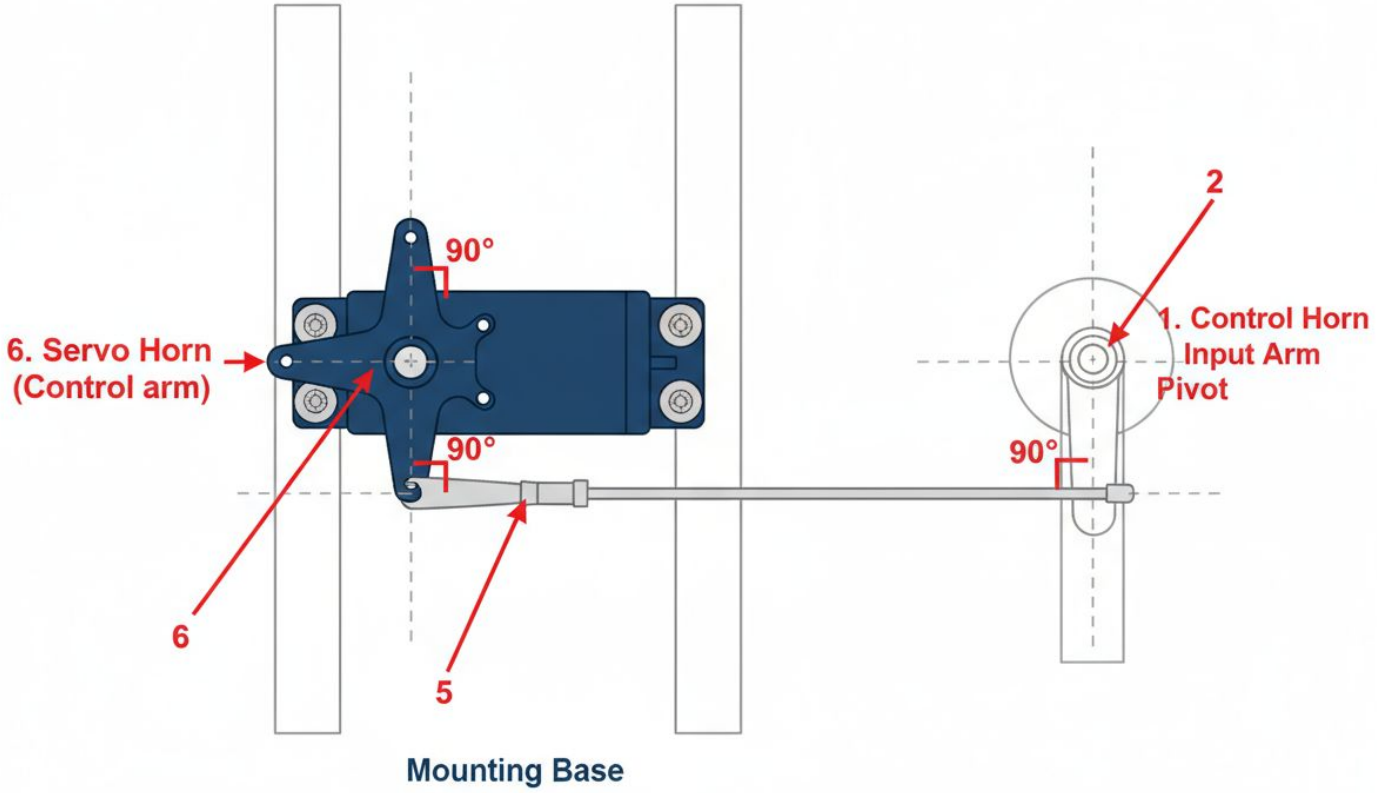
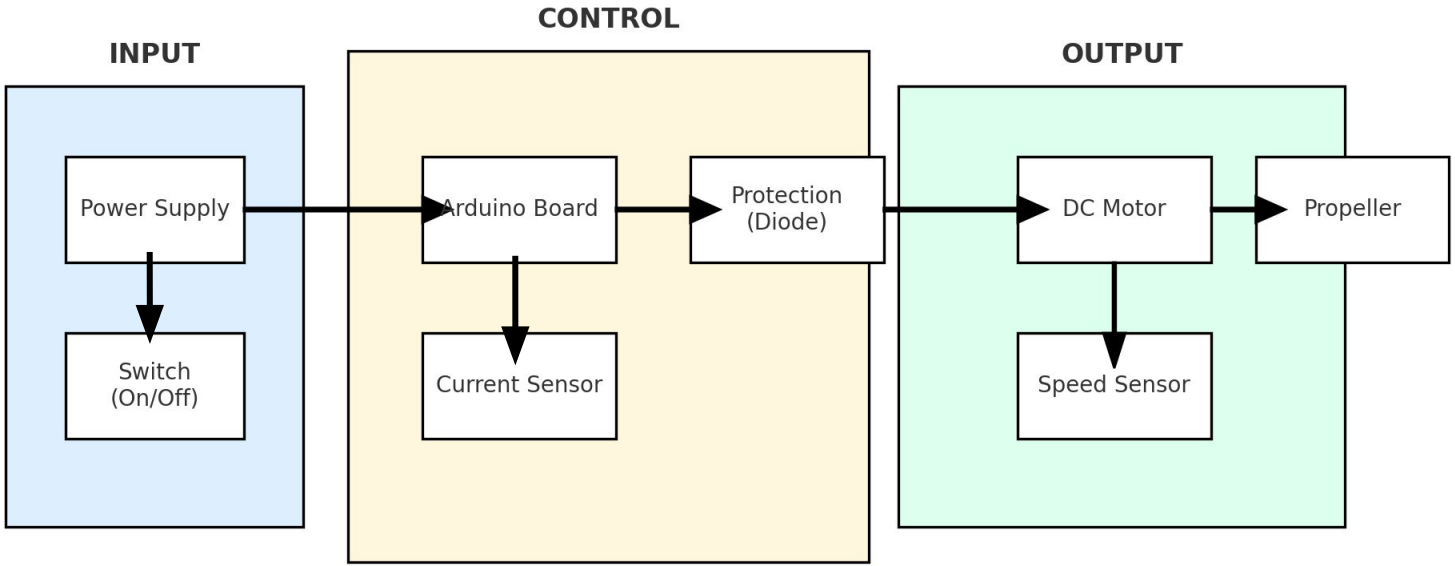
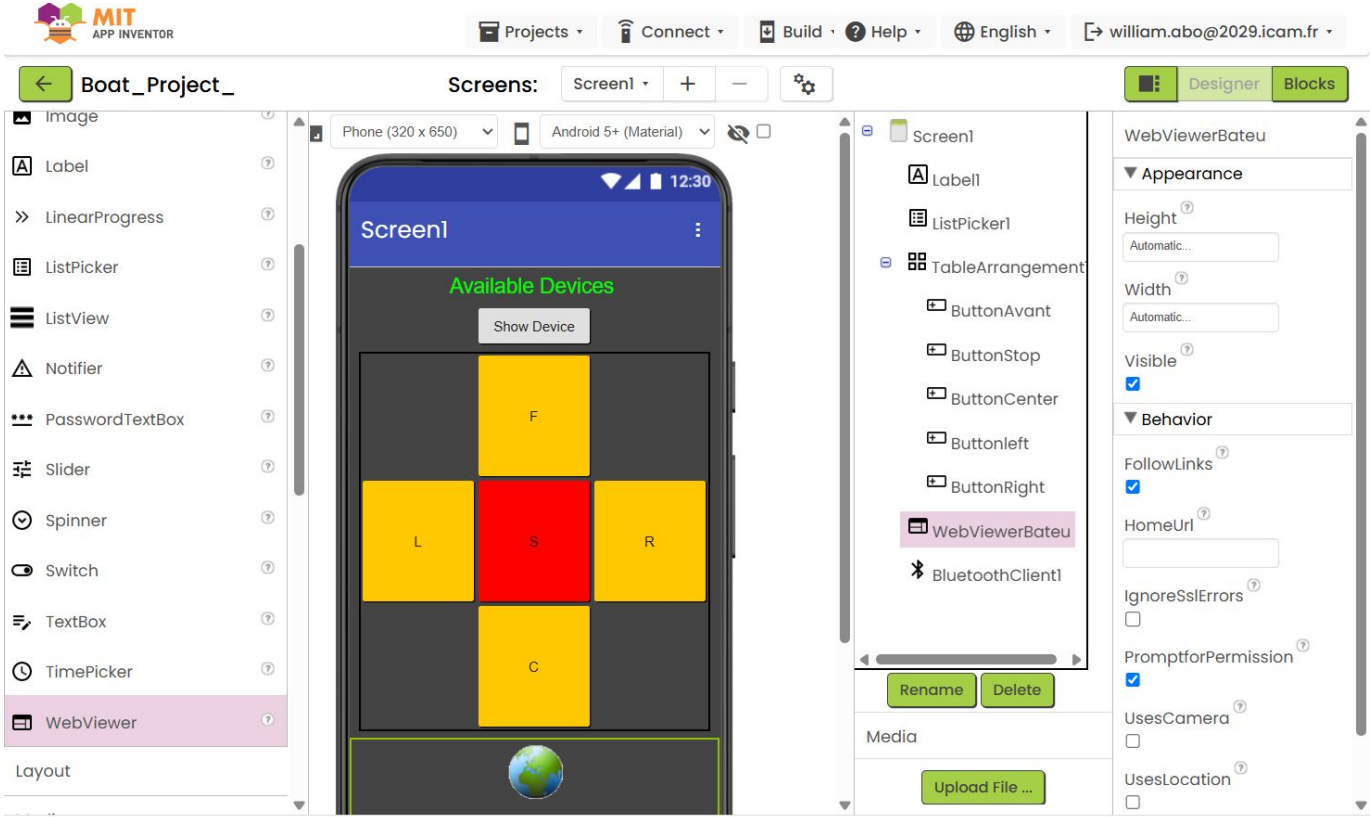


Diagram of Connections



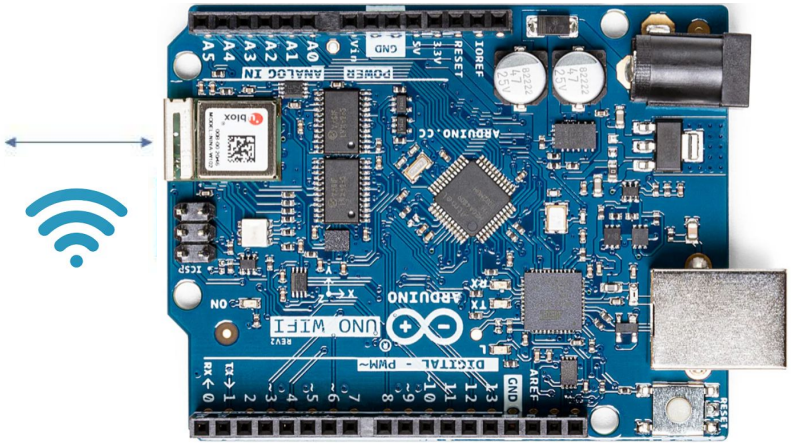
Programing

Auto-control of the boat with the App



Communication application with Arduino wifi







[Smartphone with App Inventor]
↓ (via Wi-Fi)
[SoC with integrated TCP/IP protocol stack module in the card]
↓
[Motor, servo, sensors etc.]

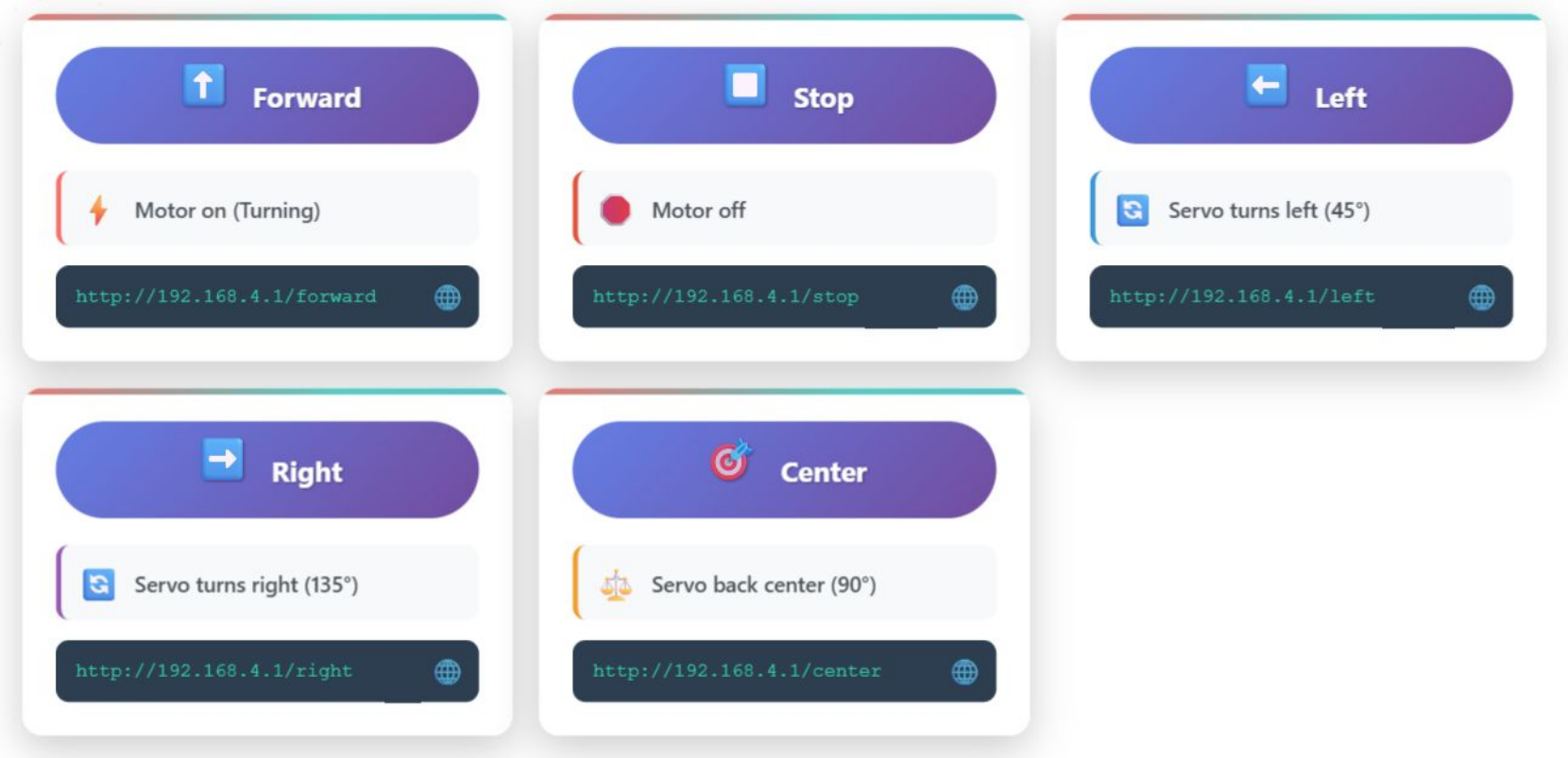
Wifi comunication system



WiFi Configuration

SSID:	"Bateau"
Password:	"12345678"
IP Address:	192.168.4.1

Controls

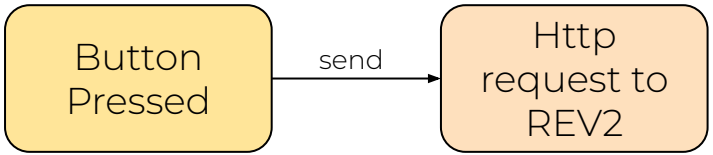

Forward
Motor on (Turning)
http://192.168.4.1/forward
Stop
Motor off
http://192.168.4.1/stop
Left
Servo turns left (45°)
http://192.168.4.1/left
Right
Servo turns right (135°)
http://192.168.4.1/right
Center
Servo back center (90°)
http://192.168.4.1/center

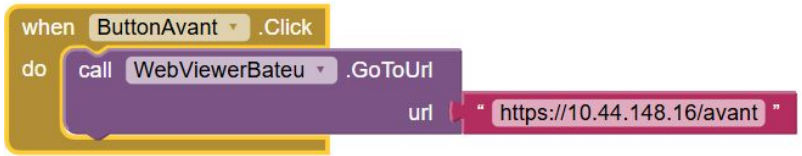
Wifi comunication system



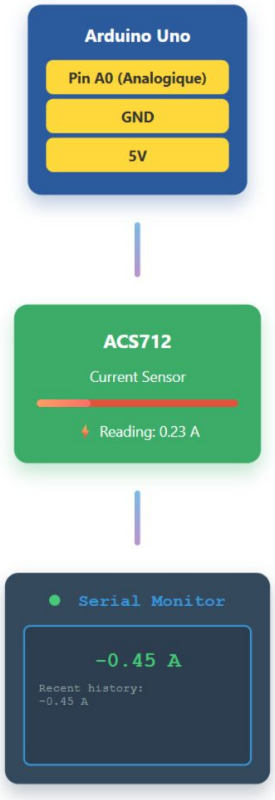
- Connection between Arduino WiFi and MIT Inventor Phone App.
- Send command to theses addresses:

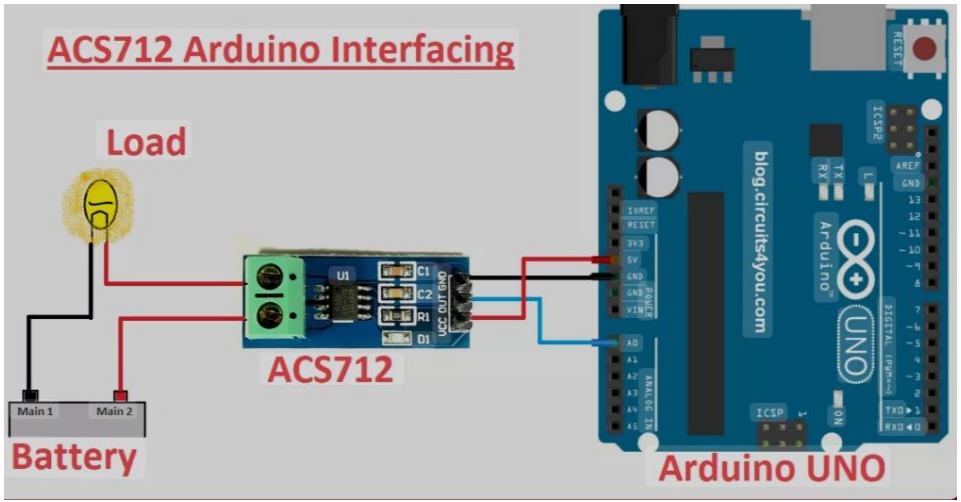
http://192.168.4.1/forward
http://192.168.4.1/stop
http://192.168.4.1/left
http://192.168.4.1/right
http://192.168.4.1/center





Current Sensor Control



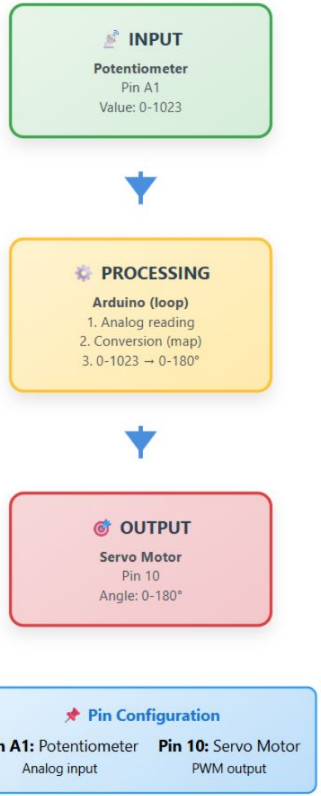


Where
Load

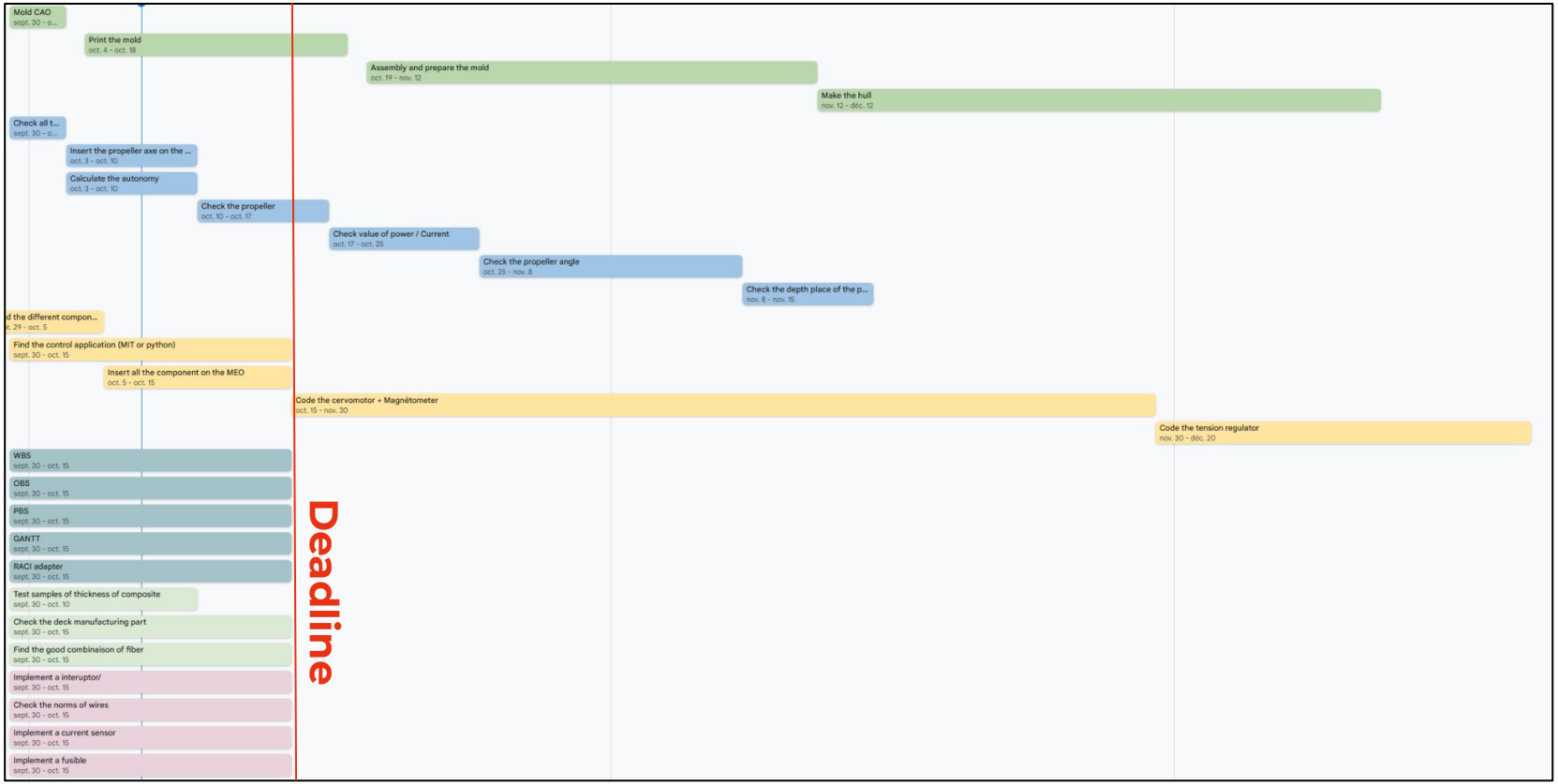
Dc Motor

Structured programs for sensor reading, motor control, communication, and error management.

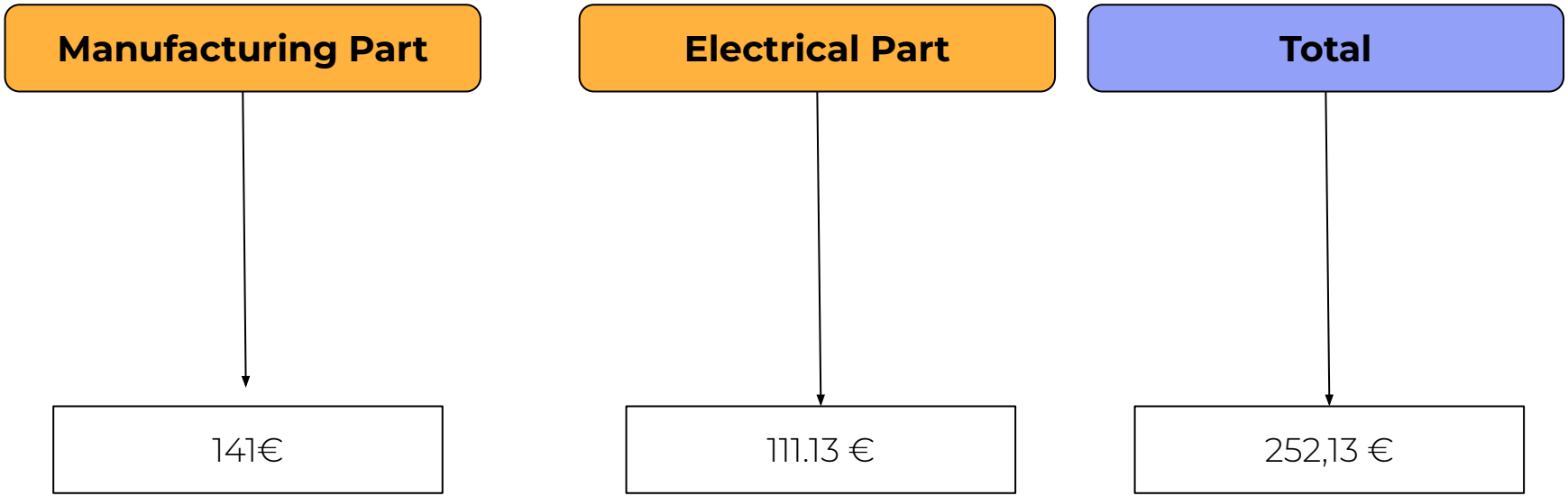
Servo motor control



Organisation - Gantt diagram



Final budget



Leisure Boat B2 Project

Thank you for listening!

Presentation made by William JUNIOR-ABO, Lucas LE GOFF,
Theo SCHLOESSER-GRIMAUD, Lilou PICHON.

APPENDICES

1. Table of Mechanical datas - Last year Reminders
2. Power Calculations
3. Energy and Information Chain
4. Wires Dimensions
5. Life Cycle Analysis
6. DashBoard Organisation

```
void loop() {
  WiFiClient client = server.available();
  if (client) {
    String requete = client.readStringUntil('\r');
    client.flush();

    if (requete.indexOf("/avant") != -1) {
      digitalWrite(IN1, HIGH);
      digitalWrite(IN2, LOW);
      analogWrite(ENA, 255); // vitesse max
      Serial.println("AVANT");
    }
    else if (requete.indexOf("/stop") != -1) {
      analogWrite(ENA, 0);  // arrêt moteur
      digitalWrite(IN1, LOW);
      digitalWrite(IN2, LOW);
      Serial.println("STOP");
    }
```

```
    else if (requete.indexOf("/gauche") != -1) {
      direction.write(45);
      Serial.println("GAUCHE");
    }
    else if (requete.indexOf("/droite") != -1) {
      direction.write(135);
      Serial.println("DROITE");
    }
    else if (requete.indexOf("/centre") != -1) {
      direction.write(90);
      Serial.println("CENTRE");
    }
```

1. Reminders:

Category	Parameter	Value
Coefficients	Cb (Block Coefficient)	*0,53*
	Cp (Prismatic Coefficient)	*0,77*
	Cm (Mid-section Coefficient)	*0,77*
	Cw (Waterline Surface Coefficient)	*0,73*
Mechanical	Theoretical Min Speed (m/s)	*0,5*
	Theoretical Max Speed (m/s)	*4,16*
	Froude Number (Fn)	*1,56*
	Friction Coefficient (Cf)	*0,0039*
	Reynolds Number	*2'553'777,8*



[13]



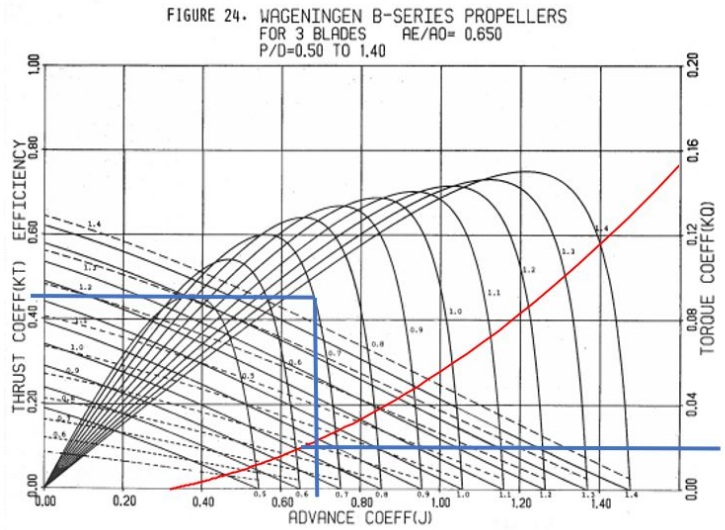
2. Final power calculation (with the propeller)

Our inputs :

Propeller				RESISTANCE	
D (m)	P (m)	P/D	AE/A0	V (m/s)	Rt (N)
0,0585	0,039	0,67	0,65	4,16	10

Graphical values and outputs :

GRAPHIC VALUES				
J	Kt	Kq	rendement	N (tr/min)
0,711	0,085	0,02	0,58	6000
n (tr/s)	omega (rad/s)	T (N)	Q (Nm)	P (W)
100	628,32	9,96	0,137	86,1

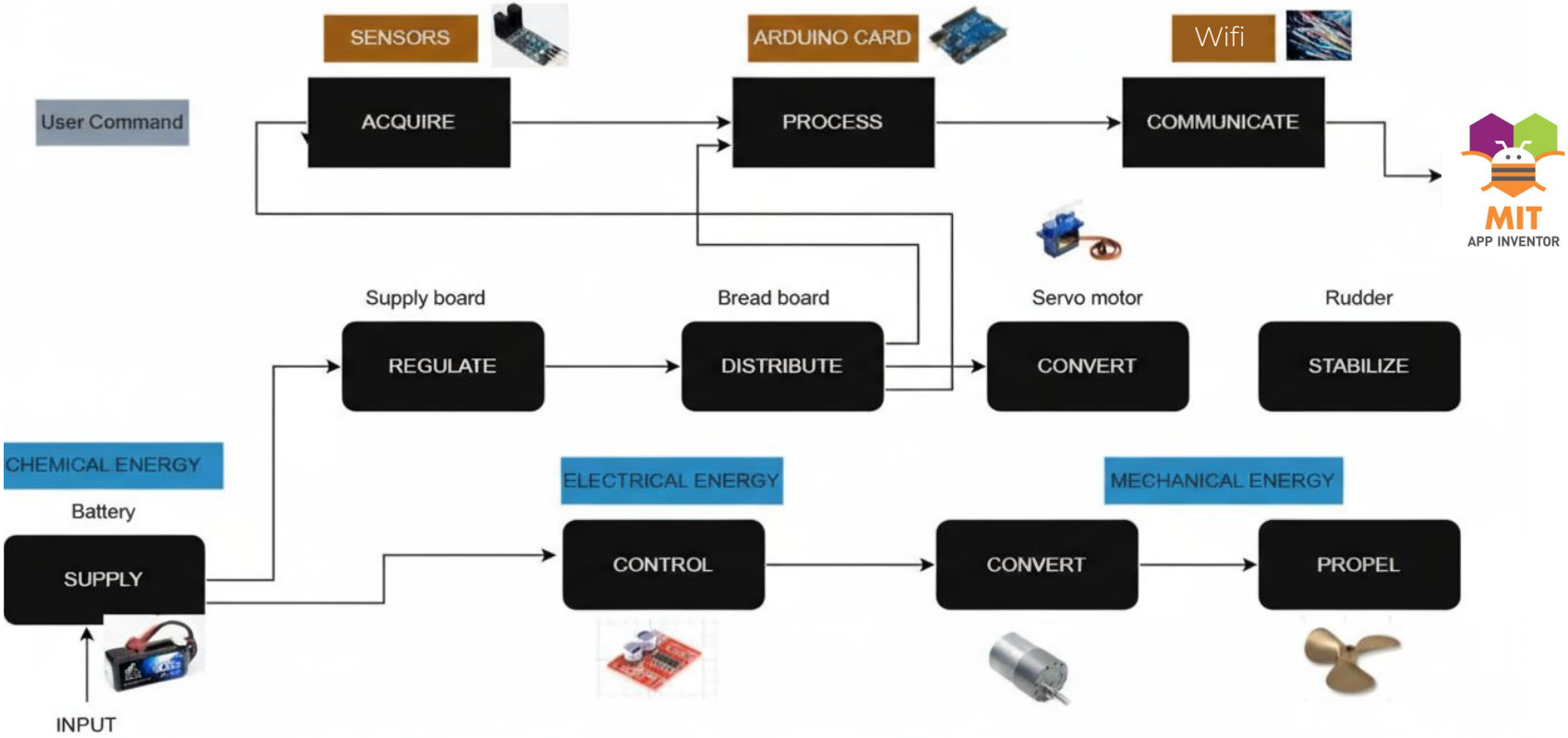


But for security, we prefer to multiply our found value by 1,2 :

P (W) = 86.1 * 1,2 = 103.32 W

Kt : Thrust coefficient
Kq : Torque coefficient
J : advance coefficient

3. Energy and Information Chain



Calculations

Motor choosed; 12 V 775 Motor (~12000 rpm)

- **No-load behavior**

The motor might spin ~12,000 rpm with no load (as rated). But once the prop is attached and pushing water, rpm will drop.

Angular speed : N=9500, ω= 2πN/ 60 = 995 rad/s

Calculate Output Torque

P= T · ω , T= P/ω

T= 100/995= 0.101N.m

Electrical input power and Current:

At 100 W mechanical output, the motor draws more than 100 W electrically (due to losses).

If efficiency η were 80% (a typical value for 775 DC motors):

η= Pout/Pin= 0.8

Pin=V · I

Efficiency for DC 775 motors is typically ~75–85%.

Assume 80 % efficiency:

Pin= 100/0.8=125 W

I=Pin/V= 125/12= 10.4A

Speed drop & margin

In practice, maybe our rpm will drop to ~8,000–10,000 rpm with load depending on drag, efficiency, prop slip .So having a motor with **no-load >/=12,000** rpm gives a safty margin.

MOTOR DRIVER AND FUSE

So we want:

- **Voltage range**: 5–15 V (safely covers $LiFePO_4$ voltage)
- **Continuous current**: ≥ 15 A
- **Peak current**: ≥ 30–40 A (safety margin)
- **PWM + direction inputs** compatible with Arduino, ESP32, Raspberry Pi, etc.

Battery + → H-bridge VCC
Battery – → H-bridge GND
Motor +/– → H-bridge OUT1/OUT2
Controller (Arduino etc.) PWM & direction pins → H-bridge inputs
Fuse ; a fuse (~15–20 A) between the battery and H-bridge for protection.
wires ; 12–14 AWG (2–3 mm^2) to minimize losses

Current sensor

Battery + → Current Sensor IN+
Sensor OUT+ → H-Bridge VCC+
Battery – → H-Bridge GND (common)
Motor connected to H-Bridge outputs
Sensor Vcc (5 V) → Arduino 5 V
Sensor GND → Arduino GND
Sensor OUT (analog) → Arduino analog pin (e.g., A0)

For ACS758-050B,

You'll read an analog voltage representing current:

I= (Vout−Vzero) / Sensitivity

- Vzero ≈ 2.5V (no current)
- Sensitivity ≈ 40 mV/A
  So if you read 3.3 V → (3.3–2.5)/0.04=20A

Wires

Wire gauge **(AWG or mm²)** based on your motor's current, wire length, and voltage (12 V).

12-14 AWG (2-3 mm²), Peak 20–25 A , to minimize heat loss

4. Wires - dimensions

Portion	Estimated length (cm)	Description
Battery → Switch	30	Direct connection, near control panel
Switch → Motor	80	Main power line to stern
Battery → H-Bridge	20	Power for control system
H-Bridge → Motor	30	Motor command circuit
Battery → Current sensor (e.g., ACS712)	20	Current measurement
Sensor → Arduino	20	Sensor signal communication
Arduino → Servo / speed sensor	20	Speed and direction circuits
Battery → LED indicator	20	Power for status display

Total cable length:
190 cm

5. LCA-Environmental impact assessment

Design phase impacts

Environmental aspect	Environmental impact	Type
Material selection and research	Use electricity + data centers	Input flow

Use phase impacts

Environmental Aspect	Environmental Impact	Type
Electricity consumption (charging)	Power consumption	Input flow

Manufacturing phase impacts

Environmental Aspect	Environmental Impact	Type
Motor/ Battery production	Metal extraction (Cu,Li)	Output flow

End-of-life phase impacts

Environmental Aspect	Environmental Impact	Type
Disassembly	Generation of fiberglass dust	Output flow

5. LCA-Results and interpretations

What are the most impactful stages of the life cycle of our boat ?



→ **Manufacturing phase**, high carbon footprint of epoxy resin production.
→ **End-of-life phase**, energy-intensive pyrolysis and waste transport.

How to reduce these impacts ?

→ To reduce impacts, : **lower-impact bio-based** resins,
minimizing material waste,
manual assembly techniques over energy-intensive tools.

6. Dashboard

Task	Start	End	Progression	People	
Hull					
Mold CAO	30/09/2025	03/10/2025	☐	Theo	Selecti…
Print the mold	04/10/2025	18/10/2025	☐	Theo	Lucas
Assembly and prepare the mold	19/10/2025	12/11/2025	☐	Lilou	Lucas
Make the hull	12/11/2025	12/12/2025	☐	Lucas	Wiliam
Propulsion					
Check all the calculations (change of the mass)	30/09/2025	03/10/2025	☐	Lilou	Selecti…
Insert the propeller axe on the MEO	03/10/2025	10/10/2025	☐	Lilou	Selecti…
Calculate the autonomy	03/10/2025	10/10/2025	☐	Wiliam	Selecti…
Check the propeller	10/10/2025	17/10/2025	☐	Lilou	Theo
Check value of power / Current	17/10/2025	25/10/2025	☐	Lucas	Selecti…
Check the propeller angle	25/10/2025	08/11/2025	☑	Lilou	Selecti…
Check the adaptation propeller/shaft	08/10/2025	15/10/2025	☐	Lucas	Selecti…
Check the depth place of the propeller	08/11/2025	15/11/2025	☐	Theo	Selecti…
MIA					
Find the different component	29/09/2025	05/10/2025	☐	Lilou	Theo
Find the control application (MIT or python)	30/09/2025	15/10/2025	☐	Wiliam	Theo
Insert all the component on the MEO	05/10/2025	15/10/2025	☐	Wiliam	Theo
Code the cervomotor + Magnétometer	15/10/2025	30/11/2025	☐	Wiliam	Theo
Code the tension regulator	30/11/2025	20/12/2025	☐	Wiliam	Selecti…
Code the application	20/12/2025	01/03/2025	☐	Wiliam	Selecti…
MEO					
WBS	30/09/2025	15/10/2025	☐	Lilou	Lucas
OBS	30/09/2025	15/10/2025	☐	Lucas	Theo
PBS	30/09/2025	15/10/2025	☐	Wiliam	Theo
GANTT	30/09/2025	15/10/2025	☐	Lilou	Wiliam
RACI adapter	30/09/2025	15/10/2025	☐	Lucas	Lilou
MATERIAL					
Test samples of thickness of composite	30/09/2025	10/10/2025	☐	Lucas	Lilou
Check the deck manufacturing part	30/09/2025	15/10/2025	☐	Lucas	Lilou
Find the good combinaison of fiber	30/09/2025	15/10/2025	☐	Lucas	Lilou
EEE					
Implement a interuptor/	30/09/2025	15/10/2025	☐	Wiliam	Selecti…
Check the norms of wires	30/09/2025	15/10/2025	☐	Wiliam	Selecti…

